



Course Project: Student Performance & Lifestyle Factors Study

American University of Ras Al Khaimah

CSAI 453 – Data Visualization

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1. Introduction

In this project, we aim to understand what drives student academic performance by analyzing both structural differences and behavioral factors. Rather than treating performance as a single outcome, we consider it as the result of multiple interacting elements such as demographics, study habits, and lifestyle patterns.

Our analysis is structured as a visual story in three stages. First, we identify who performs differently by examining factors like socio-economic status, gender, and attendance. Second, we explore why these differences occur by analyzing behavioral variables such as stress, sleep, lifestyle, and engagement. Finally, we combine these insights to highlight the key drivers of student success, making the findings both clear and actionable.

2. Dataset Preparation

We created two datasets by integrating relevant features from multiple sources and applying preprocessing and feature engineering techniques to ensure consistency and usability.

2.1 Dataset for Dashboard 1:

This dataset focuses on identifying performance differences across student groups.

Key Features:

- Gender, Age, SES, Parental Education
- Study Hours, Absences
- G1, G2, G3 (Grades)

Derived Features:

- $\text{Attendance} = 1 - (\text{Absences} / \text{Max Absences})$
- Attendance Category = Low (<0.75), Medium (<0.85), High
- $\text{Final Score} = (G1 + G2 + G3) / 3$
- Performance = Low (<10), Medium (<15), High

2.2 Dataset for Dashboard 2:

This dataset explains performance using behavioral factors. All numeric values were normalized for consistency.

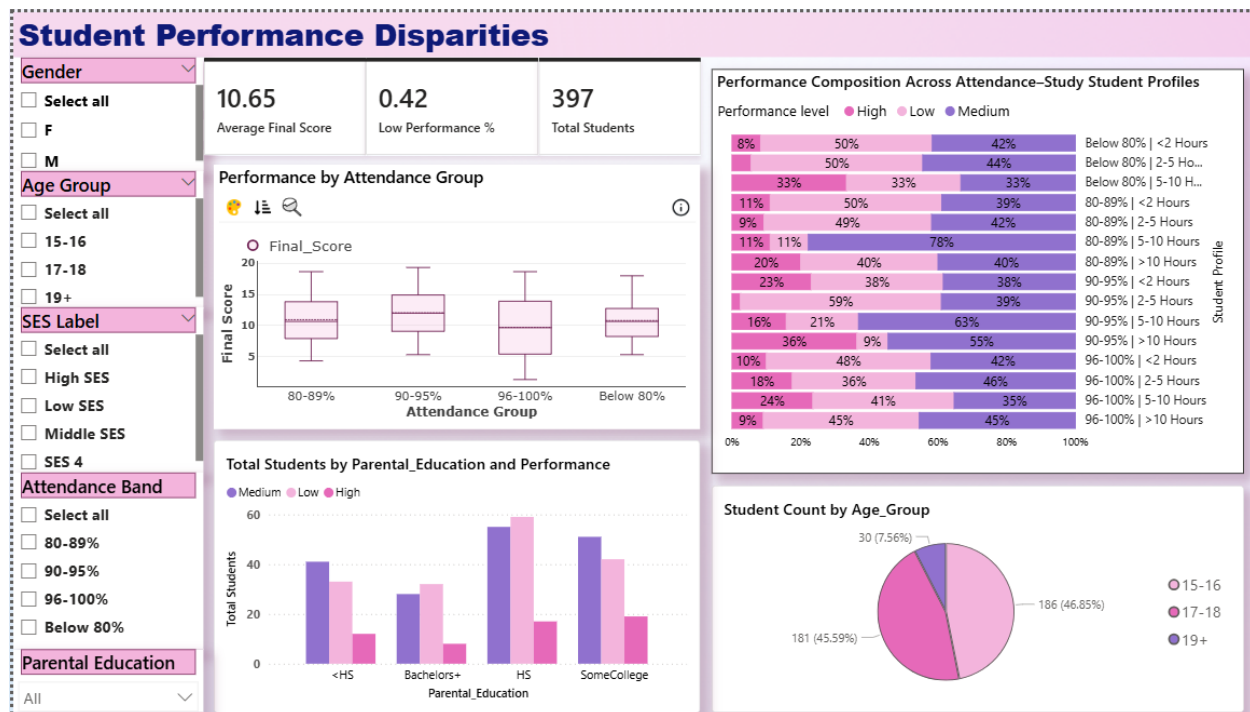
Selected Features:

- Study Time, Free Time, Social Activity, Alcohol Consumption
- Stress Level, Sleep Hours, Weekly Study Hours
- Attendance (%), Extracurricular Activities, Final Score

Derived Features:

- Lifestyle Score = avg(goout, freetime, alcohol)
- Study Intensity = avg(study time, weekly study)
- Engagement Score = avg(attendance, extracurricular)
- Stress Category = Low (<4), Medium (<7), High
- Performance = Low (<50), Medium (<75), High

3. Dashboard 1:



Below are the details of this dashboard

What: Data	Student-level tabular data were used. The dataset included demographic, academic engagement, and performance attributes: Gender, Age, Study_Hours, Attendance, Parental_Education, SES, Final_Score, and Performance.
What: Derived	<i>Several derived attributes were created to support clearer segmentation and comparison. Attendance_Band was derived from Attendance using grouped categories: Below 80%, 80–89%, 90–95%, and 96–100%. Age_Group was derived from Age. SES_Label was derived from SES. Student_Profile was created from Attendance_Band and Study_Hours. Low_Performance_Flag was derived from Performance. Profile_Sort_Key was added for logical ordering of student profiles.</i>
What: KPI / Measures	Summary measures were created in Power BI to support dashboard interpretation. These included Total Students, Average Final Score, Low Performers, Low Performance %, and Student Count. The KPI cards displayed in the dashboard were Average Final Score, Low Performance %, and Total Students.
Why: Story	Student performance disparities were examined to identify which student groups were performing differently. A descriptive, segmentation-oriented perspective was adopted to reveal at-risk profiles and unstable attendance-based performance patterns before causal explanations were explored in Dashboard 2.
Why: Tasks	Performance disparities were explored across student groups, student segments were compared, academically at-risk profiles were identified, variation in final scores across attendance levels was examined, and detail-on-demand was supported through filtering and lookup.
Question 1	Question: Which student profiles are most likely to fall into low academic performance when attendance regularity and study-time commitment are considered together? Founded Answer: Greater academic risk was concentrated in profiles characterized by weaker attendance and lower study commitment. In contrast, stronger attendance–study combinations were associated with larger shares of medium- and high-performing students. Encode: 100% Stacked Bar Chart Title: At-Risk Student Profiles Marks: segments Channels: color hue for Performance Level, horizontal position for Student_Profile, vertical extent for percentage of students. Idiom choice: A composition-based view was required so that the proportions of low, medium, and high performers within each student profile could be compared fairly. Since each bar was normalized to 100%, differences in group size did not distort comparisons across profiles. Sorting / Ordering: Dashboard
Question 2	Question: Which attendance groups show the widest variation and lowest median in student performance? Founded Answer: Lower-attendance groups were associated with lower median scores and wider variation, while higher-attendance groups appeared more stable and stronger in performance. Encode: Box Plot Title: Performance Distribution by Attendance Marks: box-and-whisker marks Channels: horizontal position for Attendance_Band, vertical position for Final_Score, box and whisker length for spread, quartiles, and range. Idiom choice: A distribution-based view was required so that not only average performance, but also median, variability, and instability could be examined across attendance groups. This made it possible for inequality in outcomes to be observed more clearly.

Question 3	Question: How are students distributed across different age groups? Founded Answer: The majority of students were concentrated in the 15–16 and 17–18 age groups, while the 19+ category represented a much smaller portion of the dataset. Encode: Pie Chart Title: Student Count by Age Group Marks: Segments Channels: Angular size and area for proportion of students, color hue for Age_Group. Idiom choice: A composition-based view was required to represent how the total student population is divided across age groups. The pie chart makes proportional differences immediately visible and allows dominant and underrepresented categories to be identified at a glance.
Question 4	Question: Are there noticeable differences in performance between students with different parental education levels, and which group tends to perform better overall? Founded Answer: Performance differences across parental education levels were present but not strongly pronounced. All groups contained students across low, medium, and high performance, although high school and some college categories showed slightly higher counts of medium and high performers. Encode: Clustered Column Chart Title: Student Count by Parental Education and Performance Marks: Bars Channels: Horizontal position for Parental_Education, vertical height for Student Count, color hue for Performance Level. Idiom choice: A comparison-based view was required to examine differences in performance distribution within each parental education group. Grouped bars allow direct comparison across performance categories and make variations between groups clear without ambiguity.
How: Facets / Views	Juxtaposed coordinated views were used. One view was used for profile-based performance composition, and one view was used for attendance-based score distribution. KPI cards were added to provide summary context for both visuals.
How: Reduce	Complexity was reduced through grouping and categorization. Continuous attendance values were reduced into Attendance_Band, age values were reduced into Age_Group, socio-economic coding was reduced into SES_Label, and attendance plus study hours were reduced into a single segmentation field, Student_Profile.
Aggregation	In the first visual, student records were aggregated by Student_Profile and Performance Level, and the percentage of students in each category was summarized. In the second visual, Final_Score values were aggregated by Attendance_Band through median, quartiles, spread, and outliers.
Manipulate	Navigation and filtering were supported through slicers and selection controls. Detail-on-demand was supported through tooltips.
Controls	Interactive slicers were used for Gender, Age_Group, SES_Label, Attendance_Band, and Parental_Education. Checkbox-style filtering and direct selection were used to refine the visible student population.
Views: Dashboard Role	The dashboard provides a demographic perspective by showing age distribution and comparing performance across parental education levels, helping users interpret results within the population context.
Summary Contribution	The dashboard shows that most students are mid-adolescents and that parental education has only a limited influence on performance, adding context to overall student analysis.

4. Dashboard 2:



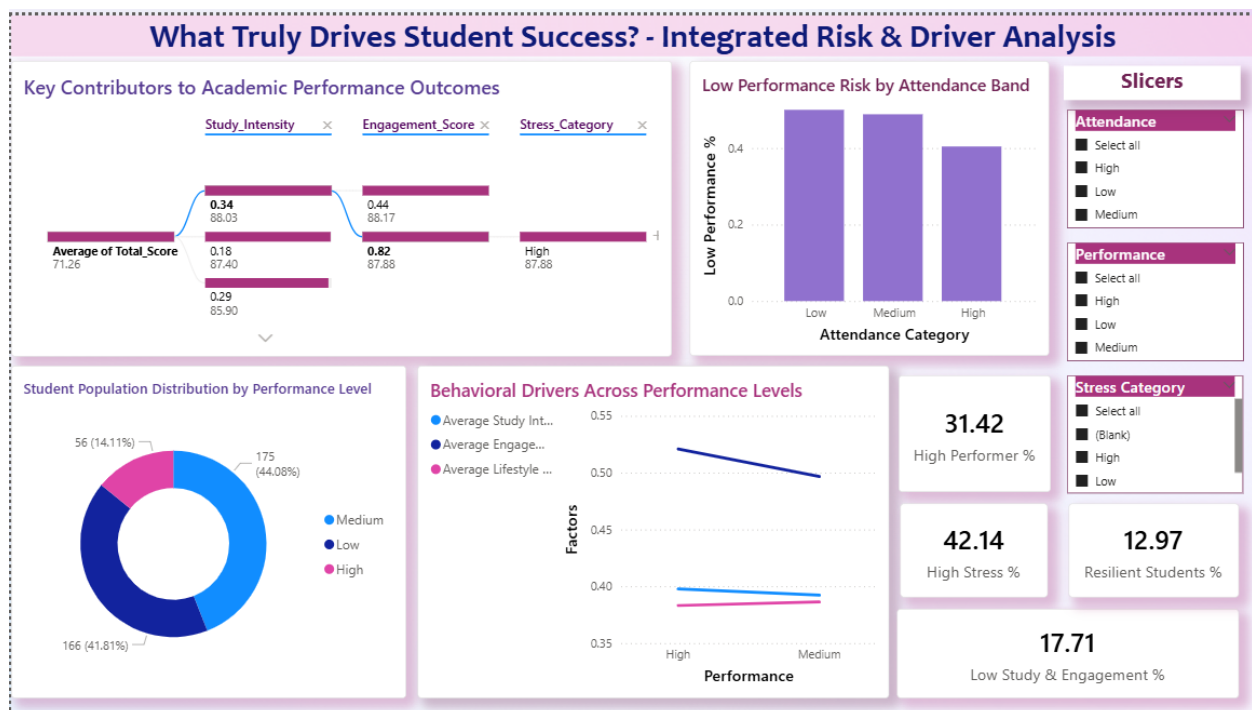
Below are the details of this dashboard

What: Data	The second dashboard examines why these differences occur. It dives into students' habits and lifestyle, such as stress, sleep, social life, and study intensity. This helps us understand the reasons for the patterns we observed in the first dashboard.
What: Derived	Numeric features were normalized so they are on the same scale. These included study_time, go_out, free_time, D_alc, W_alc, Stress_Level, Sleep_Hours_per_Night, Study_Hours_per_Week, and Attendance (%). Lifestyle_Score came from the attributes goout_norm, freetime_norm, Dalc_norm, and Walc_norm. Study_Intensity was based on the attributes Study_Hours_norm and Study_Hours_per_Week_norm. Engagement_Score was calculated using Attendance_norm and Extracurricular_numeric. Stress_Category was defined using: =IF(Stress_Level<4,"Low",IF(Stress_Level<7,"Medium","High")). Performance was defined using: =IF(Final_Score<50,"Low",IF(Final_Score<75,"Medium","High")).
What: KPI / Measures	Summary measures shown in the dashboard included Average Stress Level, Average Sleep Hours, and Average Attendance.
Why: Story	Student performance differences were looked at from an lifestyle point of view. The dashboard shows why some students do better by looking at stress, sleep, social life, study intensity and engagement habits. This helps understand why some students perform better than others.
Why: Tasks	Lifestyle impact on performance was explored, medium and high performers were compared, major habits affecting student results were identified, and users were able to filter different student conditions interactively.

Question 1	Question: How lifestyle behaviour influence performance? Founded Answer: High performers maintain higher average scores than medium performers across all lifestyle score levels. As lifestyle score increases, high performance shows a weak positive relationship, while medium performance shows a weak negative relationship after an earlier peak. Encode: Line Chart. Title: Lifestyle Score vs Academic Performance. Marks: Lines with markers. Channels: Horizontal position for Lifestyle Score, vertical position for Average Total Score, colour hue for Performance groups (High, Medium). Idiom choice: A trend-based view was required to compare score changes across lifestyle levels between performance groups.
Question 2	Question: What habits or factors affect student performance the most? Founded Answer: The strongest factors affecting medium and high performance were engagement score, stress levels, and sleep hours. Higher performers showed better sleeping hours, more stress levels, and stronger engagement score. Encode: Clustered Bar Chart. Title: Academic and Wellness Factors Vs Performance. Marks: Segments. Channels: Horizontal position for Performance category, vertical height for normalized score, and color hue for each factor. Idiom choice: A comparison-based view was required so multiple continuous factors could be compared fairly across performance groups.
Question 3	Question: Is there a trade-off between social life (Lifestyle_Score), stress, and academic success(Total_Score)? Founded answer: No strong evidence of a clear trade-off between them since the data points are widely dispersed across the lifestyle-score range, Total_Score range, and the stress categories, which indicates that students with higher lifestyle scores have variable stress levels and do not consistently achieve lower or higher academic scores. Encoding: Scatter plot Title: Life Style VS. Academic Success VS. Stress Marks: points Channels: Vertical position for Total_Score (Numerical), Horizontal position for the Lifestyle_Score (Numerical), Color Hue for the stress categories (Categorical), and size for the Stress_level (Numerical) Idiom choice: A scatter plot was used to examine the relationship between lifestyle score and academic success, with stress categories.
Question 4	Question: Which factor has the strongest impact on performance? Founded answer: The Lifestyle_Score and Study_Intensity show the strongest relationship, with a correlation of about -0.01. This means there is a weak negative relationship between them. As one increases, the other slightly decreases. Overall, the correlations are very small, so no factor appears to have a strong impact on performance. Encoding: Heatmap Title: Correlation Heatmap Marks: points Channels: Horizontal and vertical positions represent variables, and color intensity encodes the strength and direction of correlation between them. Idiom choice: A heatmap was used to show how variables are related to each other and their relationships, making it easy to see strong and weak correlations with performance (Total_Score).

How: Facets / Views	Juxtaposed coordinated views were used. One view presents the relationship between Lifestyle_Score and academic performance using a line chart. Another view compares academic and wellness factors using a clustered bar chart. KPI cards were placed between visuals for summary context. And a view to represent the correlation between the variables using a heatmap, while another view represents the relationships between the academic success, lifestyle, and stress using a scatter plot
How: Reduce	Complexity was reduced through normalization, grouping, and feature engineering. Multiple raw variables were reduced into Lifestyle_Score, Study_Intensity, and Engagement_Score. Stress_Level was reduced into Stress_Category. Final_Score was reduced into Performance categories.
Aggregation	In the first visual, records were aggregated by Lifestyle_Score and Performance using average total score. In the second visual, factors were aggregated by Performance category using average normalized values. KPI cards used overall averages. In the scatter plot, points with the same Lifestyle_Score and stress category values were aggregated by averaging Total_Score.
Manipulate	Navigation and filtering were supported through slicers, sliders, and tooltips. Users could inspect relationships and compare selected groups dynamically.
Controls	Interactive slicers were used for Activities outside academics, Study_Intensity, Stress_Category, Engagement_Score, and Performance. Range sliders and checkbox filtering were used to refine visible students.
Views: Dashboard Role	The dashboard provides an explanatory perspective by connecting student performance with behaviour, wellness, and school engagement patterns.
Summary Contribution	The dashboard shows that higher performers generally maintain stronger scores across lifestyle levels, while engagement, sleep, and stress are major factors that distinguish medium and high performers. The final data structure used was: Final_Score, Performance, Study_Intensity, Lifestyle_Score, Stress_Level, Stress_Category, Sleep_Hours, Engagement_Score.

5. Dashboard 3:

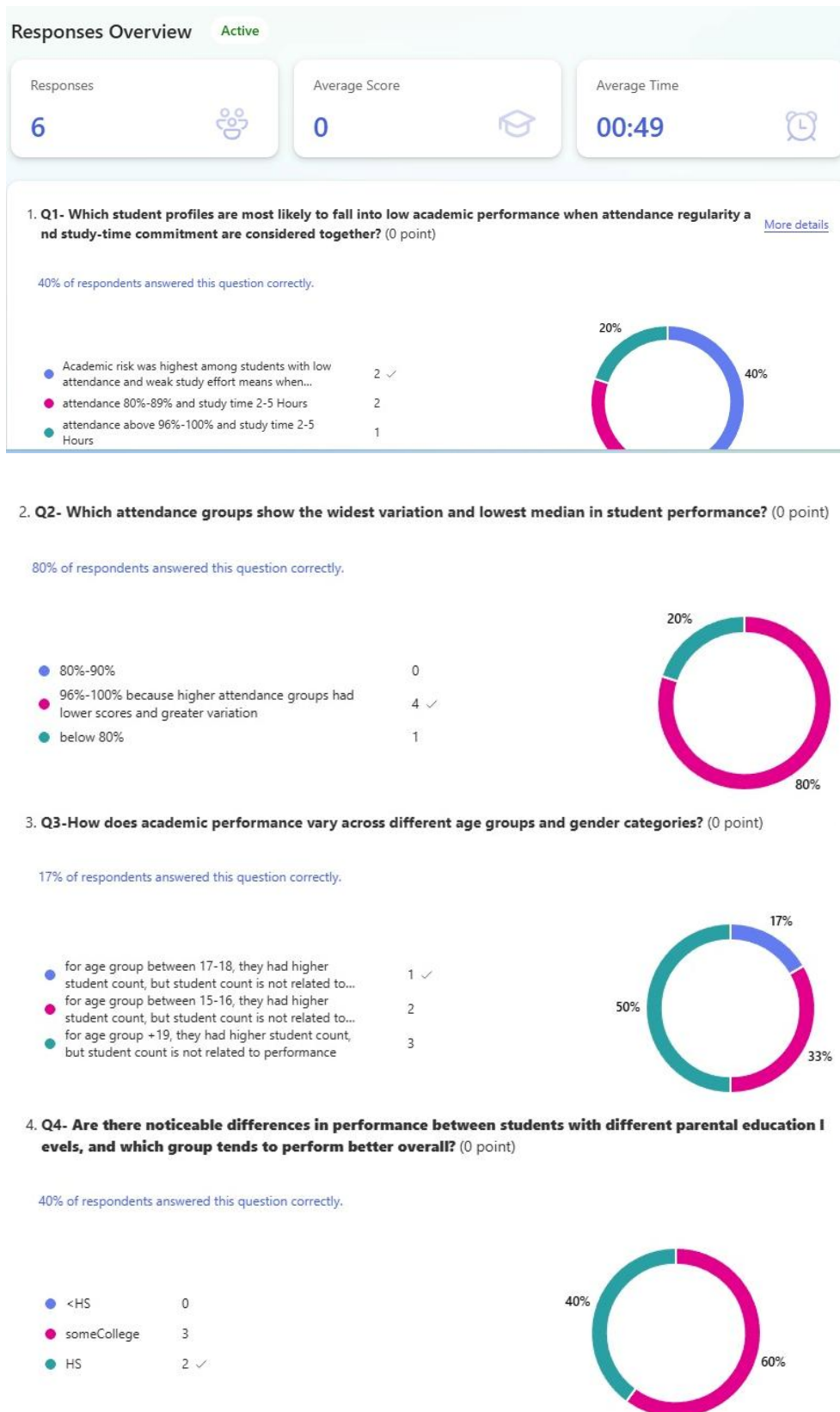


Below are the details of this dashboard

What: Data	Student-level tabular data were used, integrating academic outcomes, attendance regularity, and behavioral attributes. The dataset included Performance, Final_Score, Attendance_Category, Study_Intensity, Engagement_Score, Stress_Category, and Lifestyle_Score.
What: Derived	<i>Resilient_Students, Low_Study_Engagement, and other summary indicators were derived for KPI representation.</i>
What: KPI / Measures	Summary measures were created in Power BI to support high-level interpretation. These included High Performer %, High Stress %, Resilient Students %, and Low Study & Engagement %. KPI cards displayed these measures to provide immediate contextual understanding of success, stress prevalence, resilience, and academic risk.
Why: Story	The summary dashboard integrates structural risk patterns and behavioral drivers to answer the overarching question of what most strongly drives student success. Rather than introducing new segmentation, this stage synthesizes findings from the first two dashboards and highlights priority factors and actionable insights.
Why: Tasks	Key performance drivers were ranked, attendance-related risk was summarized, behavioral differences across performance levels were compared, resilience under stress was examined, and headline indicators were provided to support decision-oriented interpretation.
Question 1	Question: What factors most strongly contribute to academic performance outcomes when behavioral drivers are considered together? Founded Answer: Study intensity was identified as the strongest contributor to performance, followed by engagement level. Stress influenced outcomes as a moderating factor rather than a primary driver. Encode: Decomposition Tree Title: Key Contributors to Academic Performance Outcomes Marks: Nodes Channels: Hierarchical position for explanatory priority, node values for relative contribution to average final score. Idiom choice: A hierarchical decomposition idiom was required to support prioritization and allow users to explore how different behavioral factors contribute to performance outcomes.
Question 2	Question: Which attendance groups face the highest risk of low academic performance? Founded Answer: Students in the low attendance category exhibited the highest risk of low performance, while students with high attendance showed substantially lower risk. Encode: Column Chart Title: Low Performance Risk by Attendance Band Marks: Bars Channels: Horizontal position for Attendance_Category, vertical height for Low Performance %. Idiom choice: A ranking-based view was required to allow direct comparison of academic risk across attendance categories without introducing distributional complexity.
Question 3	Question: How is the student population distributed across academic performance levels? Founded Answer: Most students fall within the medium and low performance categories, while a smaller proportion achieve high academic performance. Encode: Donut Chart Title: Student Population Distribution by Performance Level Marks: Segments Channels: Angular size for proportion of students, color hue for Performance category.

	Idiom choice: A composition-based idiom was selected to provide immediate population-level context and show the relative size of performance groups at a glance.
Question 4	Question: How do key behavioral drivers vary across different performance levels? Founded Answer: Study intensity and engagement scores were higher among high-
	performing students and declined with lower performance levels, while lifestyle score showed comparatively smaller variation. Encode: Line Chart Title: Behavioral Drivers Across Performance Levels Marks: Lines with point markers Channels: Horizontal position for Performance level, vertical position for average driver value, color hue for driver type. Idiom choice: A comparative trend idiom was used to highlight differences in behavioral patterns across ordered performance categories.
How: Facets / Views	Coordinated views were used to integrate prioritization, risk comparison, population context, and behavioral differentiation. KPI cards were included alongside analytical views to provide summary indicators that support rapid interpretation.
How: Reduce	Complexity was reduced by aggregating behavioral metrics into composite driver scores, categorizing attendance into discrete bands, and summarizing performance patterns through high-level KPIs rather than detailed distributions.
Aggregation	Performance outcomes were aggregated using averages and percentages. Attendance-based risk was summarized as low-performance rates. Behavioral drivers were aggregated as average normalized scores within performance groups.
Manipulate	Interactive slicers and direct selection allow users to filter by Attendance_Category, Performance, and Stress_Category. KPI cards and visuals update dynamically to support exploratory analysis and scenario testing.
Controls	Interactive slicers were provided for Attendance, Performance, and Stress_Category.
Views: Dashboard Role	The summary dashboard serves as the final synthesis layer, integrating segmentation and behavioral analysis to highlight key drivers, risk patterns, and priorities for intervention.
Summary Contribution	The dashboard demonstrates that attendance structures academic risk, while study intensity and engagement are the most influential drivers of success. It also highlights the presence of resilient students who maintain high performance under stress, translating detailed analysis into actionable insight.

6. Evaluation:



5. Q5-How lifestyle behavior influence performance? (0 point)

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50% of respondents answered this question correctly.

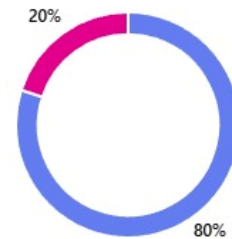
- High performers maintained higher scores at all lifestyle levels and showed a slight negative trend 2
- medium performers maintained higher scores at all lifestyle levels and showed a slight positive trend 1
- High performers maintained higher scores at all lifestyle levels and showed a slight positive trend 3 ✓



6. Q6- what's the most habits or factors that affects student performance? (0 point)

80% of respondents answered this question correctly.

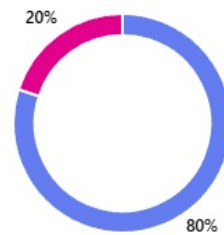
- engagement, stress level, and sleep hours 4 ✓
- engagement, and lifestyle score 1
- only stress level 0



7. Q7: Is there a trade-off between social life, stress, and academic success? (0 point)

80% of respondents answered this question correctly.

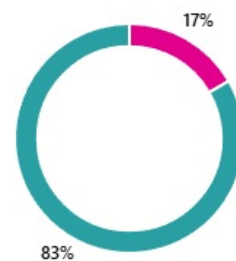
- there is no clear trade-off because academic scores remain consistent across all lifestyle levels 4 ✓
- when social life increases, academic success decreases 1
- when social life and stress decreases, academic success decreases 0



8. Q8: Which factor has the strongest impact on performance? (0 point)

83% of respondents answered this question correctly.

- study intensity 0
- lifestyle score 1
- sleep hours 5 ✓



To evaluate the clarity and effectiveness of the dashboards, a small user study was conducted with 6 students. Participants were asked to answer visualization-based questions for both Dashboard 1 and 2, and their response time and accuracy were recorded.

Qualitative feedback indicated that most visualizations were easy to follow, though the scatter plot was considered confusing by multiple participants. As future work, this visualization could be redesigned using an alternative encoding to improve interpretability.

7. Conclusion:

This project explored student performance by combining demographic and behavioral factors into a structured visual analysis. The results show that while background characteristics such as socio-economic status influence outcomes, behavioral factors like study intensity, engagement, and stress play a more significant role in explaining performance differences.

The comparative evaluation highlighted that behavior-focused visualizations were more effective in communicating insights than segmentation-based ones. Overall, the project demonstrates that meaningful data integration and thoughtful visualization design are essential for producing clear, actionable insights into student success.